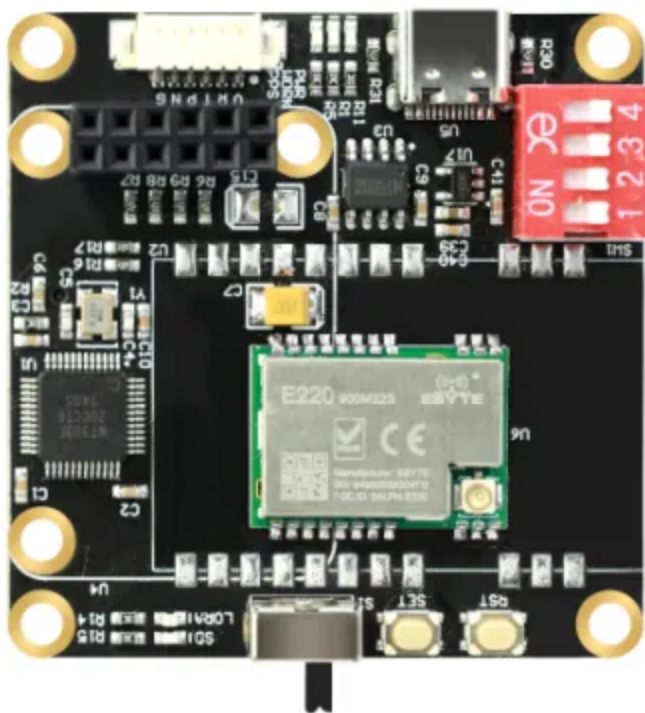


RTK-LORA评估板使用说明V1.0

- 1. 产品功能
- 2. 操作说明
 - 2.1. 指示灯说明
 - 2.2. 拨动开关说明
 - 2.3. 按键说明
 - 2.4. 端口管脚说明
- 3. 配置说明
 - 3.1. 基站
 - 3.2. 移动站
- 4. 数据来源说明
- 5. SD卡数据查看
 - 5.1. 基站的数据
 - 5.2. 移动站的数据

1. 产品功能

两块评估板连接WTRTK-982/ WTRTK-980/ WTRTK-960等RTK模块，不需要千寻等网络账号以及流量，通过LORA进行远距离无线技术传输，速度更快。两块评估板之间进行数据传输，精准定位，达到厘米级精度。同时支持SD卡离线存储原始数据。



(发射功率：22dB)



(发射功率：30dB)

2. 操作说明

2.1. 指示灯说明



PWR：模块电源灯

PPS：直接连接RTK模块的PPS引脚，定位状态指示灯（详情可参考RTK模块）

WORK：工作状态指示灯

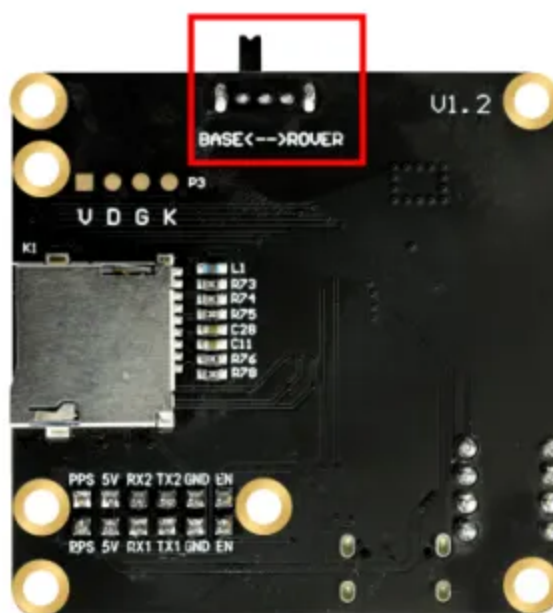
SD：SD卡状态指示灯

LORA：数据传输状态指示灯

现象	PWR	PPS	WORK	SD	LORA
亮	通电				基站：正在发送 移动站：正在接收
灭	未通电或电压不足				基站：无发送 移动站：无接收
亮1s灭1s			未定到位（仅支持移动站）		

1s闪烁1次		定定位	单点定位（仅支持移动站）	工作正常	
1s闪烁2次			差分模式（仅支持移动站）需要输出GxRMC和GxGGA这两包数据	未插入或卡损坏	
1s闪烁5次 (快闪)			正在设置		

2.2. 拨动开关说明

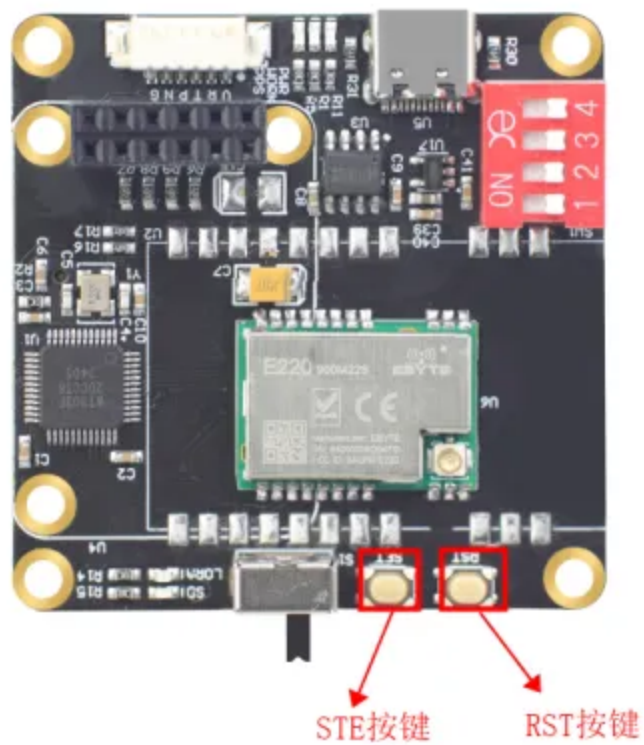


拨到BASE是基站，拨到ROVER是移动站。

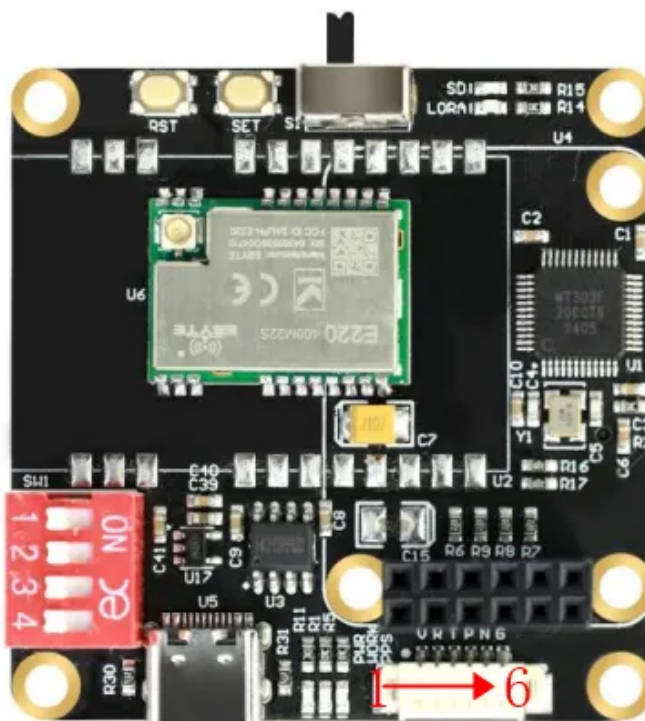
2.3. 按键说明

SET:快速配置RTK（基站/移动站）

RST:复位(重新开始工作)



2.4. 端口管脚说明



管脚序号	1	2	3	4	5	6
描述	5V电源输入	直连RTK 串口-RX1	直连RTK 串口-TX1	PPS定位状态指示灯	无	接地GND

注意：二次开发时，如单片机等接线以及供电可连接此端口。通过此串口可以获取到RTK的数据以及对RTK模块进行控制。

3. 配置说明

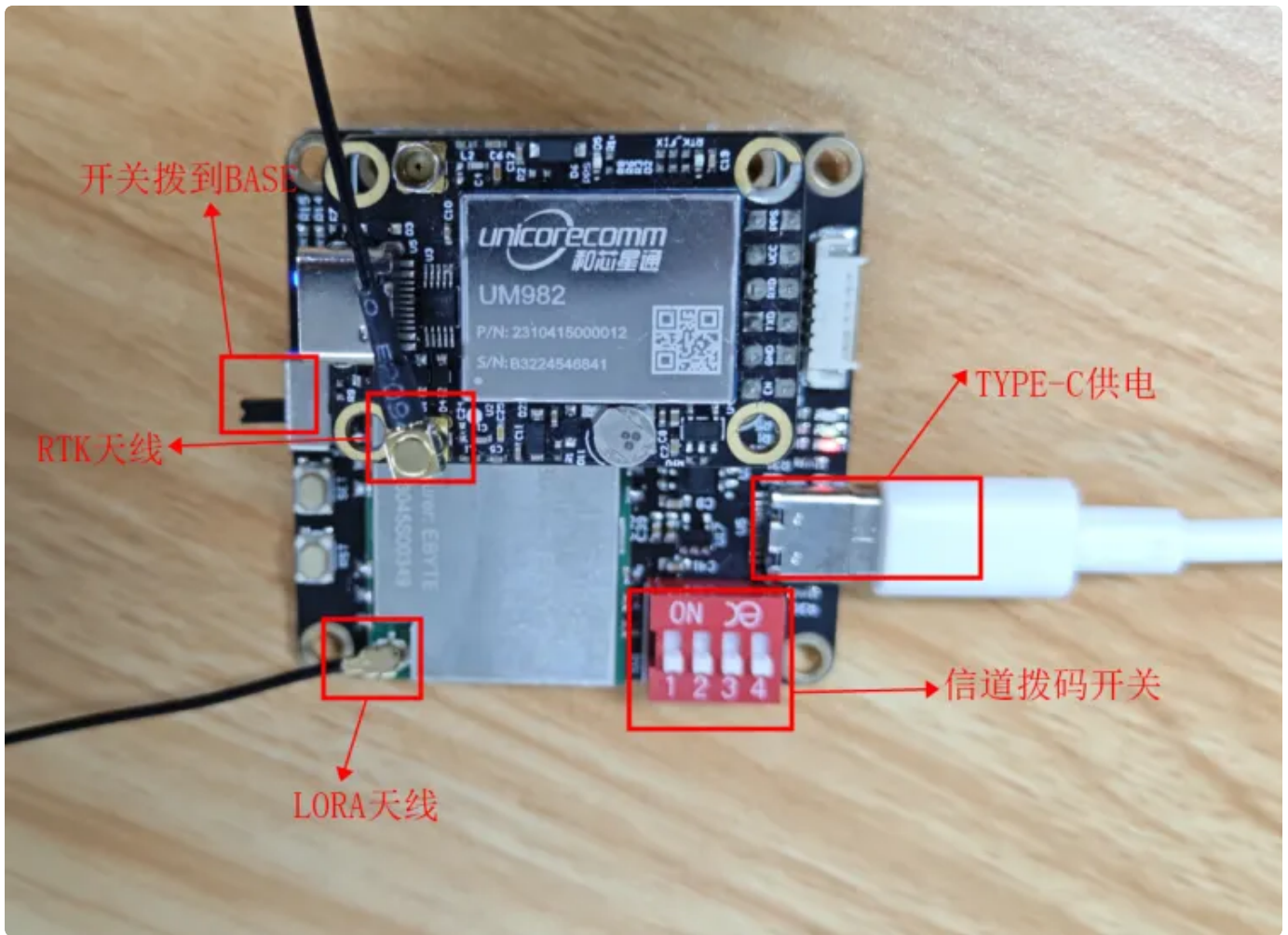
LORA天线和RTK天线不是同一种，请勿安装错。配置时保证基站和移动站的拨码开关位置一致。

3.1. 基站

- (1) 将拨片开关拨到**BASE**
- (2) 拨动**信道拨码开关**（拨到和移动站相同的位置）
- (3) 用TYPE-C给产品供**5V**的电
- (4) 长按**SET键**，直到WORK灯快闪后放开SET键，即可将RTK快速配置成**基站**。（如自己已经将RTK模块配置成基站以及其他数据之后，可以省略此步骤）

按下SET键之后会自动配置RTCM1003、RTCM1006、RTCM1074、RTCM1124、RTCM1084、RTCM1094这几包数据，如需要其他的数据可参考RTK模块的数据手册。

注意：WORK灯快闪时请勿移动整个产品。



3.2. 移动站

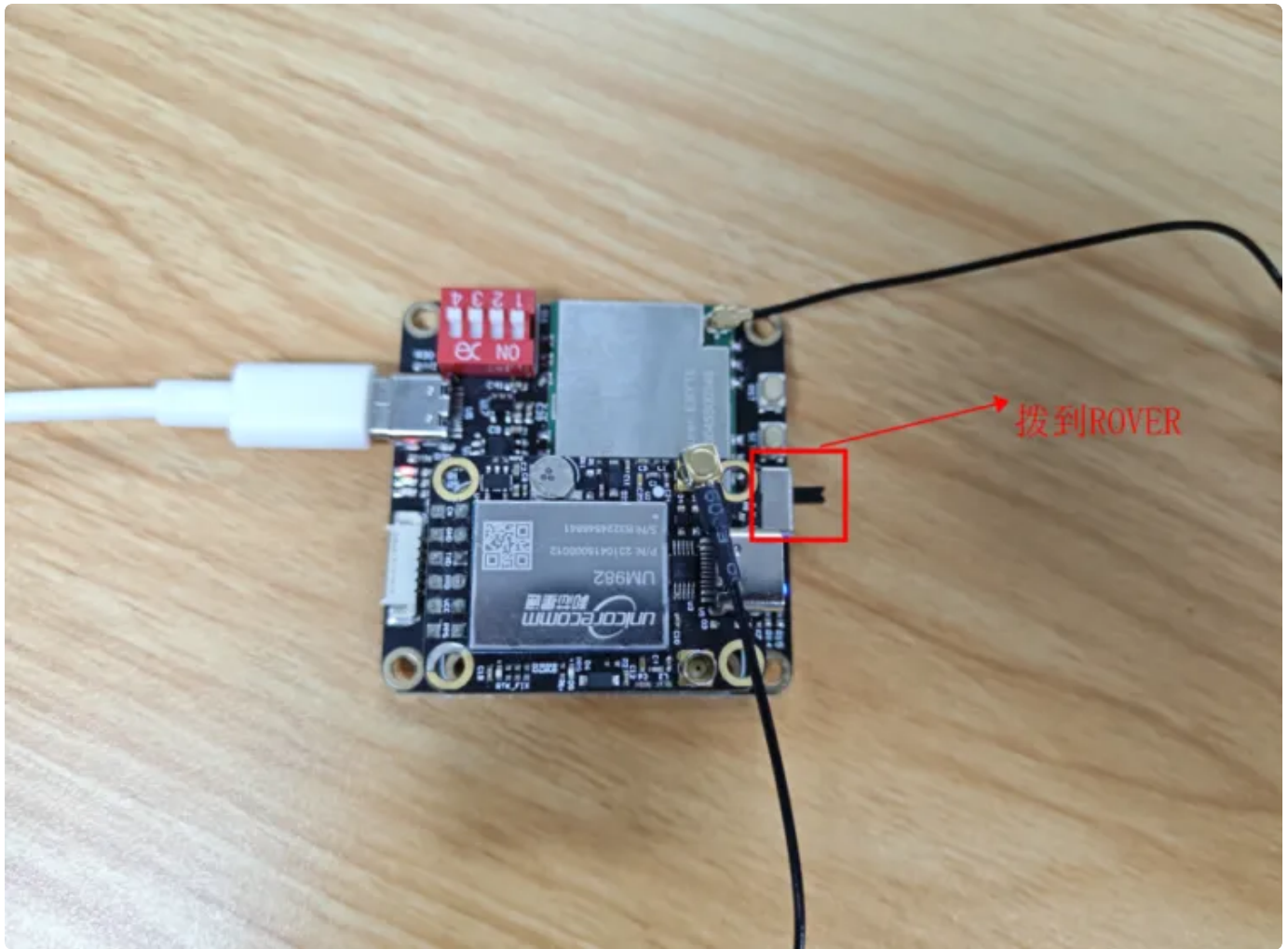
(1) 将拨片开关拨到**ROVER**

(2) 将信道拨码开关拨动成和基站**相同的位置**

(3) 用TYPE-C给产品供**5V**的电

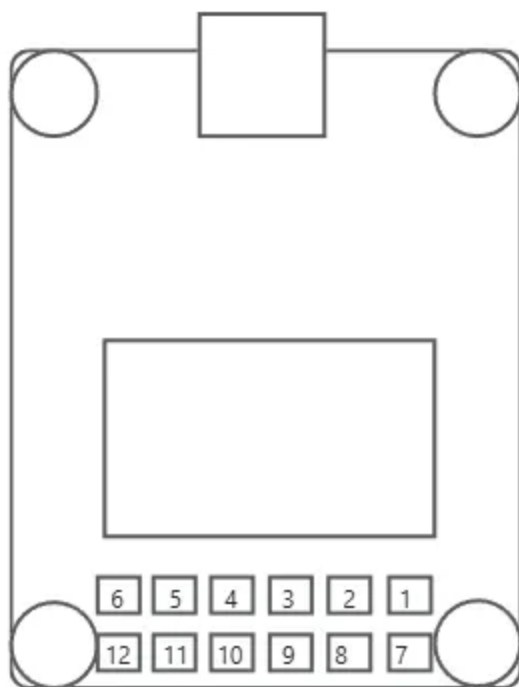
(4) 长按下**SET**键，直到WORK灯快闪后放开SET键，即可将RTK快速配置成**移动站**。（如自己已经将RTK模块配置成移动站以及其他数据之后，可以省略此步骤）

按下SET键之后会自动配置GPGGA和GPRMC这两包的数据每1s的输出，如需要其他的数据可参考RTK模块的数据手册。



4. 数据来源说明

LORA通讯以及SD的数据记录都来源RTK模块的串口2



引脚序号	引脚名称	引脚用途
1	PPS	秒脉冲
2	VCC	电源必须接5V
3	RX2	串口2数据输入
4	TX2	串口2数据输出
5	GND	地
6	EN	高电平使能脚

5. SD卡数据查看

插入SD卡后，评估板会自动将串口原始数据记录到SD卡中，可以通过读卡器查看RTK的原始信息。

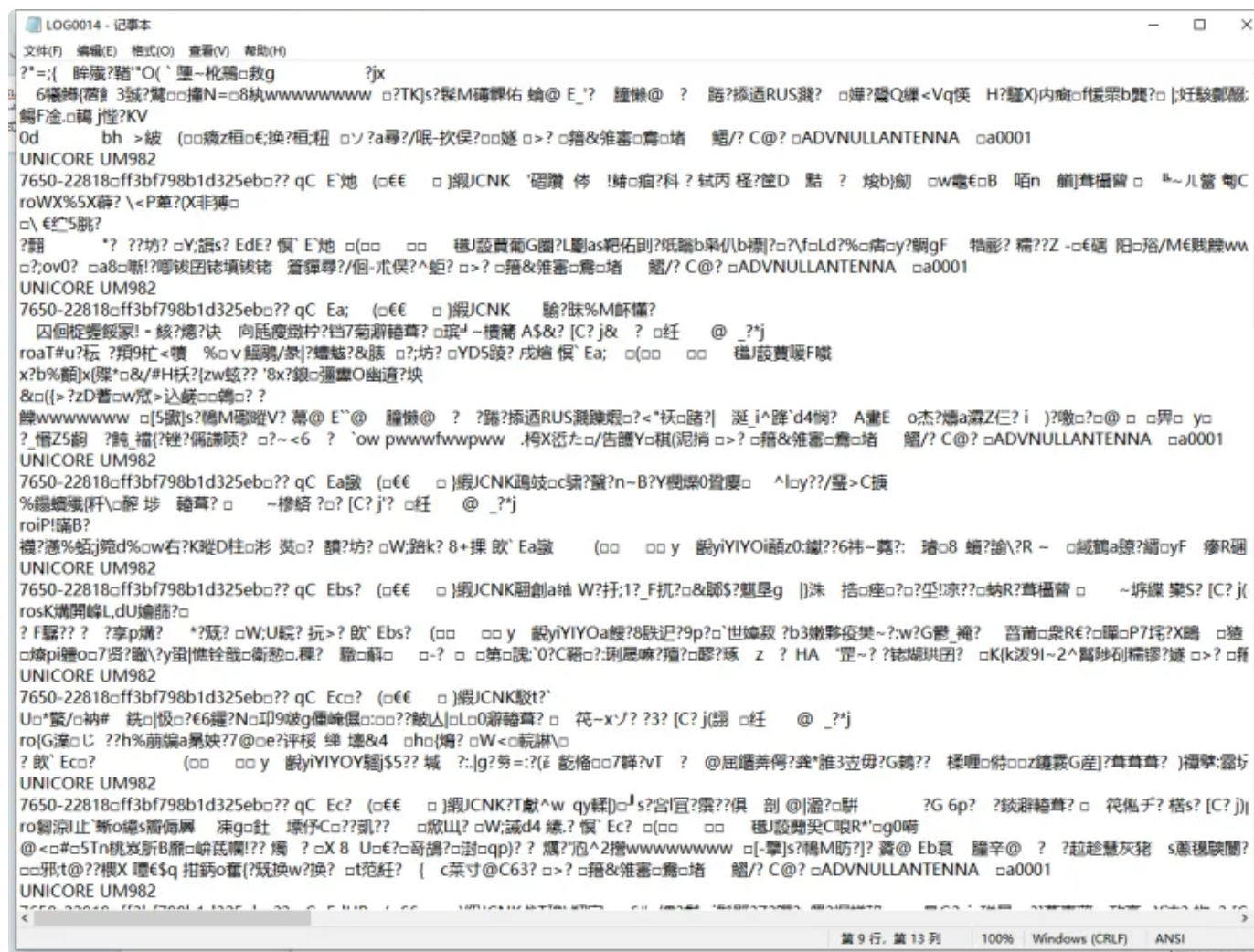
SD卡记录文件说明

此电脑 > U 盘 (D:)



日志信息：原始串口数据

5.1. 基站的数据



5.2. 移动站的数据

LOG0027 - 记事本

文件(F) 编辑(E) 格式(O) 查看(V) 帮助(H)

\$GNRMC,085139.00,A,2246.69880472,N,11351.90145498,E,0.003,248.7,140824,3.3,W,F,C*51
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